

Abstract

Understanding who is speaking within a group setting is a key component of behavioral analytics, team dynamics research, and human centered sensing systems. This project presents the design and implementation of a Bluetooth Low Energy based wearable badge system for detecting active speakers using multimodal sensor data. Each badge continuously collects sound intensity and acceleration measurements, which are transmitted to a central system and stored in a unified data format for processing and analysis. Device whitelisting is used to reduce environmental noise and ensure reliable data acquisition from known participants.

To transform raw sensor streams into structured inputs suitable for classification, the data is segmented into 20 second windows. Within each window, statistical features including minimum, maximum, mean, and standard deviation are extracted from both sound and motion signals. These features provide a compact representation of speaking behavior while preserving temporal trends. To efficiently generate labeled training data, an auto labeling approach based on heuristic sound thresholds is used to assign activity states indicating active or not active speaking periods.

These results (so far 85%) demonstrate the feasibility of privacy conscious, badge-based speaker activity recognition without recording or storing raw audio content. Potential applications include team interaction analysis, classroom engagement monitoring, workplace collaboration studies, and broader research in wearable sensing and social signal processing. This work establishes a scalable end-to-end pipeline and provides a foundation for future improvements in labeling quality, feature engineering, and real-world deployment robustness.

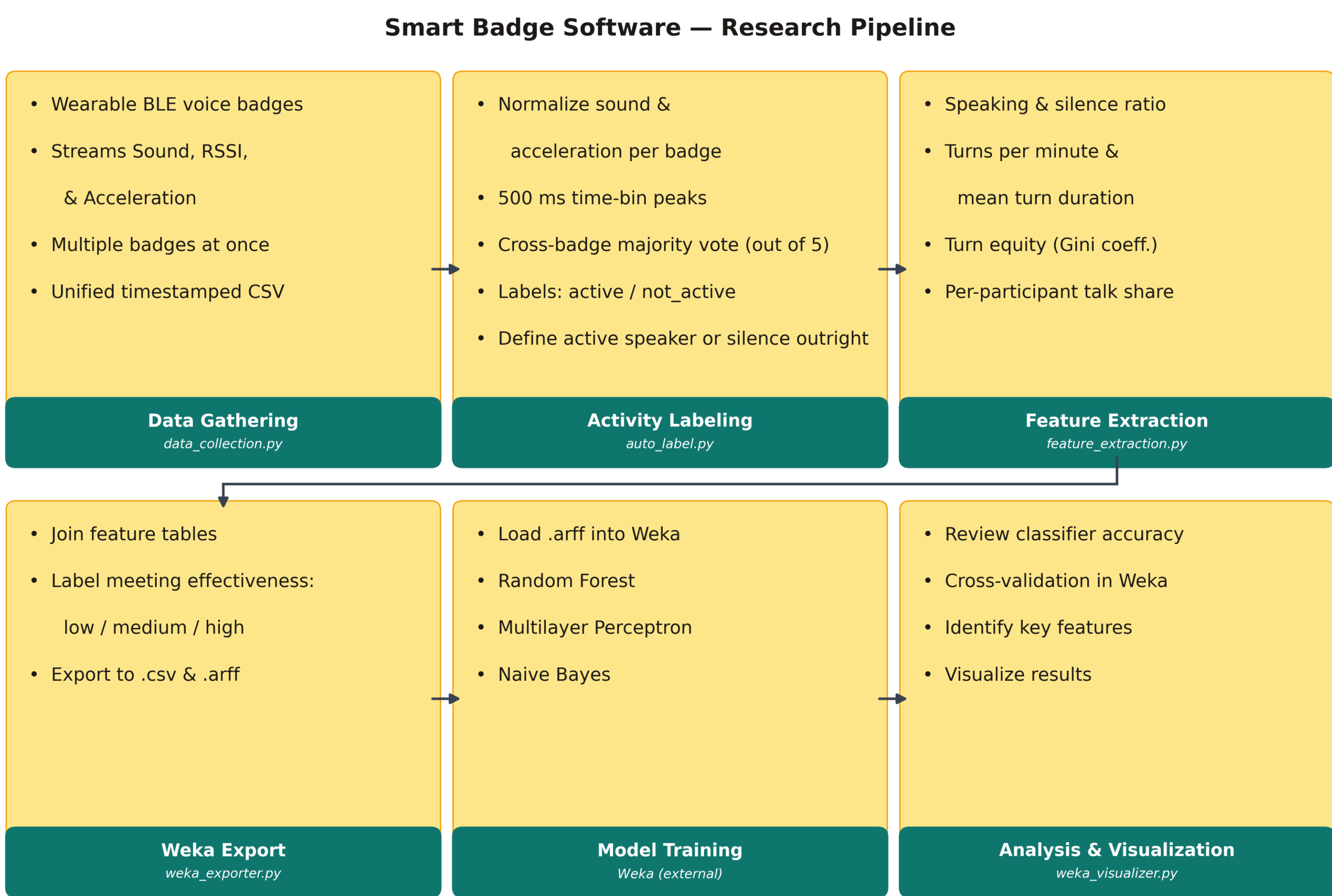
Description

- Investigated meeting productivity using **wearable voice interaction badges** worn by participants.
- Focused on **speech dynamics**, **turn-taking behavior**, and **participation equity** across group meetings.
- Used real-time BLE sensor data: **Sound Level**, **Acceleration**, and **RSSI**.
- Extracted speech activity labels per participant using peak-based cross-badge comparison in **500 ms time bins**.
- Computed meeting-level features including **speaking ratio**, **silence ratio**, **turns per minute**, and **Gini coefficient of turn equity**.
- Classified meetings into **low**, **medium**, or **high effectiveness** using self-report labels.
- Applied machine learning classifiers (Random Forest, Naive Bayes, Multilayer Perceptron) in **WEKA**.
- Found patterns linking turn-taking imbalance and silence ratio to lower meeting effectiveness.
- Proposed **automated, real-time meeting feedback** tools to improve group collaboration and inclusivity.
- Designed a system to identify meeting effectiveness **without collecting any contextual data** about the meeting

Existing Research

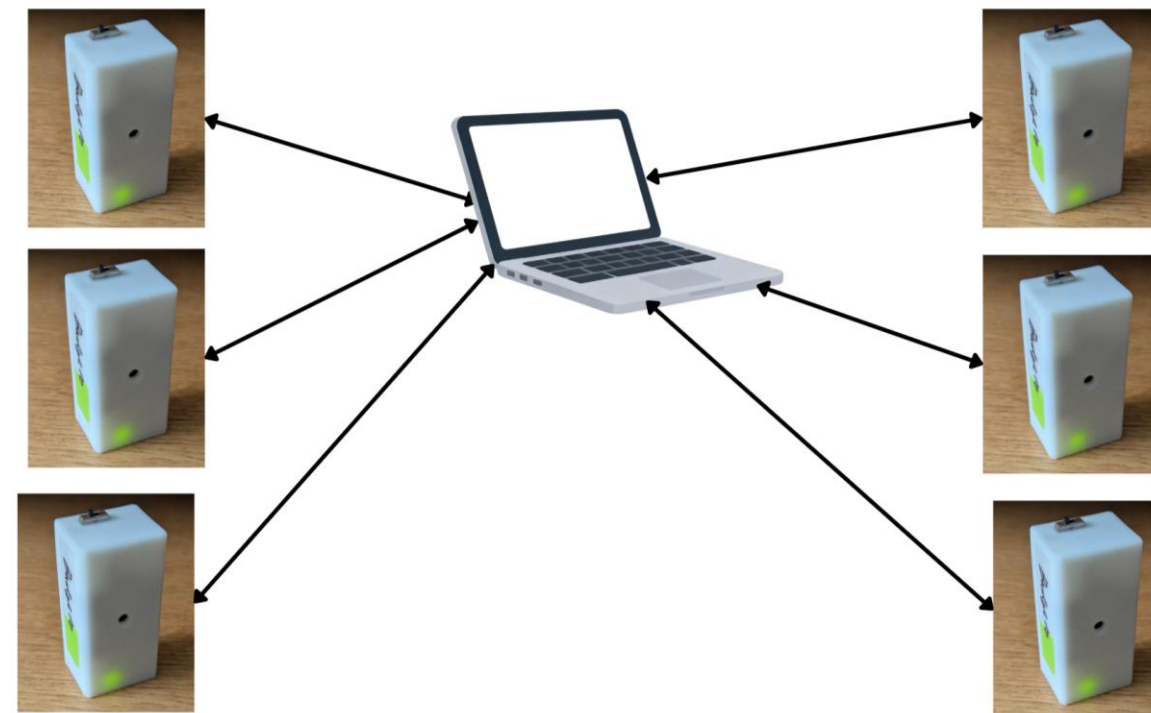
STUDY	APPROACH	ACCURACY / PERFORMANCE	LIMITATIONS
MIT Sociometric Badges	Wearable (audio + motion + IR + Bluetooth)	~87% (face-to-face interaction detection) (MIT Media Lab)	Expensive; derived metrics vary
Open Badges (2016)	Open-source wearable badges	N/A	Requires custom assembly
Validation Study (2018)	Commercial badge validation vs ground truth	~50–65% sensitivity, ~95% specificity (arXiv)	Vendor metrics unreliable
Accelerometer Speech Detection (2012)	Speech via vibration (no audio)	Not consistently reported (lower than audio methods) (Studocu)	Reduced accuracy
Smartwatch Conversation Detection (2025)	Smartwatch (audio + motion)	82%	Needs phone pairing
Multimodal Engagement (2024)	Smart glasses (video + audio + eye tracking)	96%	Privacy + cost issues
Conversing Group Detection (2018)	Accelerometers + transfer learning	~70–85% (task-dependent)	Needs multiple users
Hybrid Meeting Engagement (2025)	Physiological wristbands (HR, EDA)	N/A	Extra sensors required

Workflow



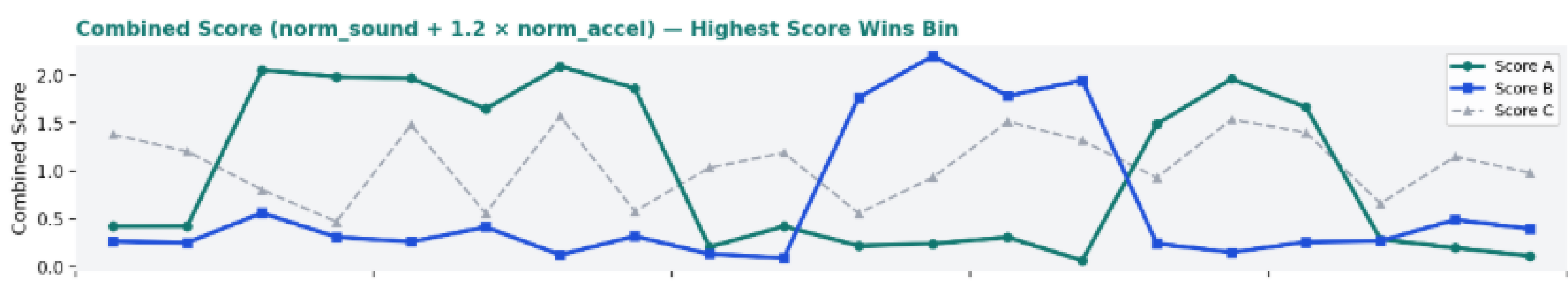
Data Collection

- Connect to Badges via **Bluetooth Low Energy** based on MAC
- System supports **~6-8 Badges** connected to one host
- Badge data is **pulled at 16 HZ**
- Data includes:
 - Sound Level**
 - Acceleration**
 - RSSI**
- This data gets stored in a CSV file



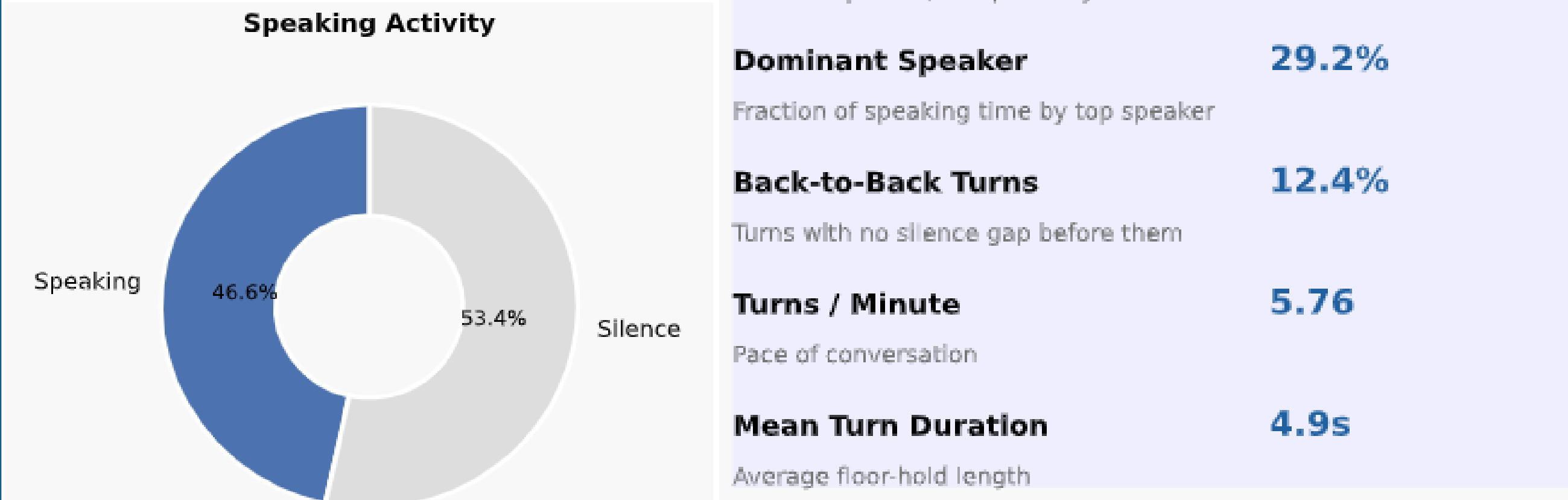
Auto Data Labeling

- Data is split into **500 ms time windows** for analysis
- Each badge is scored using **IQR-normalized** sound and accelerometer data
- A **rolling 1s mean vs. 60s baseline** threshold detects when speech is present
- Accelerometer data is weighted 1.2x** to distinguish the speaking badge from bystanders
- The highest-scoring badge **wins a majority vote across 5 windows** before being labeled active
- A post-processing pass removes **short bursts, fills brief gaps, and merges**



Feature Extraction

- The auto-labeled CSV is loaded, and raw samples are aggregated into **500ms bins per badge**
- The meeting is divided into **fixed time windows** (default 5 minutes) and the same features are computed per window to capture how dynamics shift over time
- Three output tables are produced:
 - one row per meeting
 - one row per participant
 - one row per time window



Machine Learning Results

Multilayer Perceptron (14-fold cross-validation)



Random Forest (14-fold cross-validation)



Analysis & Conclusions

- Both Random Forest and Multilayer Perceptron **achieved 92.9% accuracy** (Kappa = 0.89) across **14 meetings**
- High effectiveness meetings were **classified perfectly** by both models, suggesting a clear behavioral signature exists
- Medium effectiveness was the **hardest class to detect**, with one misclassification each per model
- Multilayer Perceptron showed better **probability calibration** (MAE 0.127 vs. 0.234) despite identical accuracy
- Turn-taking dynamics and participation balance (micro turn ratio, back-to-back ratio, speaking ratio) were the **strongest predictors** of meeting quality

Future Work

- Expand dataset beyond 14 meetings to **improve generalizability** and **reduce overfitting risk**
- Test across **remote meetings, larger teams, and professional settings**
- Build a **real-time feedback dashboard** that alerts facilitators when a meeting trends toward low effectiveness
- Increase the deployment of badges** from 6-8 to a wider amount to see if outcomes degrades or improves
- Move from WEKA to a **custom model for detection and identification**
- Attempt to get **model on-device** for faster real-time response